Ritabrata Dasgupta

Ph.D. | Geodynamics | Solid Earth Geophysics

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Research Interests

Geodynamics, Solid Earth Geophysics, Structural Geology, Tectonics

Appointments

Marie Skłodowska-Curie Action Postdoctoral Fellow 2025 – Present

Laboratoire de Géologie de Lyon Terre Planètes Environnement, Université Claude Bernard Lyon 1, Lyon, France

Host: Prof. Maëlis Arnould and Prof. Maxim Ballmer

Research area: Slab Plume Interaction in Deep Earth Region - SPIDER

Postdoctoral Researcher in Geodynamics 2022 – 2025

Department of Earth System Sciences, Yonsei University, Seoul, Republic of Korea

Host: Prof. Changyeol Lee

Research area: Geodynamic modelling of plume lithosphere interactions

UGC Junior/Senior Research Fellow and Teaching Assistant 2015 – 2022

Department of Geological Sciences, Jadavpur University, Kolkata, India

Education

Ph.D. in Geodynamics 2015 – 2022

Department of Geological Sciences, Jadavpur University, Kolkata, India

Advisor: Prof. Nibir Mandal

Research area: Mechanics of lithospheric subduction processes

M.Sc. in Applied Geology 2013 – 2015

Department of Geological Sciences, Jadavpur University, Kolkata, India

Percentage: 76.31

B.Sc. in Geological Sciences 2010 – 2013

Department of Geological Sciences, Jadavpur University, Kolkata, India

Percentage: 70.33

Publications

1. Hazarika, P. J., Baruah, A., **Dasgupta, R.**, Mandal, N. (2026). **Control of mechanical stratification on stress localization in a shallow magma plumbing system containing a pre-existing fracture: Findings from thermo-mechanical, visco-plastic models**. *Solid Earth Sciences*.
2. **Dasgupta, R.**, Lee, C. (2025). **Deciphering the underlying hydrous plume geometry and dynamic topography of intraplate volcanoes: A computational fluid dynamic model analysis**. *Physics of Fluids*.
3. Hazarika, P., **Dasgupta, R.**, Baruah, A., Mandal, N. (2024). **Ground surface displacements and stress localization driven by dual magma chamber dynamics: analytical and numerical model estimates**. *International Journal of Earth Science*.
4. **Dasgupta, R.**, Lee, C. (2024). **Lithospheric weakening by a small-scale plume and its geodynamic implications**. *Earth and Planetary Science Letters*.
5. **Dasgupta, R.**, Mandal, N. (2022). **Trench topography in subduction zones: a reflection of the plate decoupling depth**. *Frontiers in Earth Science*.
6. **Dasgupta, R.**, Mandal, N. (2022). **Role of double-subduction dynamics in the topographic evolution of the Sunda plate**. *Geophysical Journal International*.
7. **Dasgupta, R.**, Sen, J., Mandal, N. (2021). **Bending curvatures of subducting plates: old versus young slabs**. *Geophysical Journal International*.
8. **Dasgupta, R.**, Mandal, N., Lee, C. (2021). **Controls of subducting slab dip and age on the extensional versus compressional deformation in the overriding plate**. *Tectonophysics*.
9. **Dasgupta, R.**, Mandal, N. (2018). **Surface topography of the overriding plates in bivergent subduction systems: A mechanical model**. *Tectonophysics*.

Conference Presentations

1. **Dasgupta, R.**, Arnould, M., Ballmer, M., Tierny, J. (2026). **Automatic detection and time tracking of mantle plumes using topological data analysis**. Ada Lovelace Workshop, Charles University, Prague, Czech Republic.
2. **Dasgupta, R.**, Lee, C. (2024). **Geodynamic implications of small plume driven weakening of continental lithosphere**. International Geological Congress, Busan, Republic of Korea.
3. **Dasgupta, R.**, Lee, C. (2024). **Weakening of continental lithosphere by baby plume: insights from numerical models**. Japan Geoscience Union Meeting, Chiba, Japan.
4. **Dasgupta, R.**, Lee, C. (2023). **Small plume-driven weakening of the lithosphere: insights from numerical models**. American Geophysical Union Fall Meeting, San Francisco, USA.
5. **Dasgupta, R.**, Mandal, N., Lee, C. (2022). **Role of double subduction interactions on the topographic evolution of the Sunda Plate**. 7th National Conference on Rock Deformation and Structures, Banaras Hindu University, Varanasi, India.
6. **Dasgupta, R.**, Mandal, N., Lee, C. (2022). **Role of Miocene subduction interactions on the evolution of the Sunda Plate topography**. Ada Lovelace Workshop on Numerical Modelling of Mantle and Lithosphere Dynamics, ETH Zürich and Eotvos Lorand University, Hungary.
7. **Dasgupta, R.**, Sen, J., Mandal, N. (2021). **Bending curvatures of subducting plates: role of slab age**. 6th Conference on Rock

Deformation and Structures, Central University of Kerala, India.

8. **Dasgupta, R.**, Lee, C., Mandal, N. (2019). **First-order surface topography of subduction zones in numerical models: the roles of subducting slab age and dip**. European Geosciences Union General Assembly, Vienna, Austria.
9. **Dasgupta, R.**, Mandal, N. (2018). **First-order topography in bi-vergent subduction systems: the effects of slab interaction**. 5th Conference on Rock Deformation and Structures, University of Delhi, India.
10. **Dasgupta, R.**, Lee, C., Mandal, N. (2018). **Role of subducting slab dip and wedge rheology on the development of back-arc topography**. Spring Symposium of Petrological Society of Korea, Andong, Republic of Korea.
11. **Dasgupta, R.**, Nayak, S., Samanta, S. (2014). **Control of rheology on the geometry of a square boudin: an experimental approach**. 3rd Rock Deformation and Structures Conference, Dibrugarh, India.

Invited Talks

1. **A glimpse of research and teaching in Geodynamics: An overview**. IISER Tirupati, India (2025).
2. **A glimpse of research and teaching in Geosciences: An overview**. NIT Durgapur, India (2025).
3. **Subduction zones and mantle plumes: The drivers of Earth's geodynamics**. IIT Kharagpur, India (2024).
4. **Geodynamic implications of hydrous plume emplacement in continental lithosphere**. NISER Bhubaneswar, India (2024).
5. **Weakening of continental lithosphere by baby plume: insights from numerical models**. Seoul National University, Republic of Korea (2024).
6. **Subduction zones and mantle plumes: The driving engines in Earth's geodynamics**. NISER Bhubaneswar, India (2023).
7. **Miocene development of Sunda plate topography**. Yonsei Frontiers Seminar, Republic of Korea (2022).
8. **Subduction zone geodynamic modelling: an overview**. NGRI Hyderabad, India (2021).

Awards and Honors

- Marie Skłodowska Curie Postdoctoral Fellowship, MSCA Call 2024 (2025)
- Seal of Excellence, MSCA Call 2023 (2024)
- Postdoctoral Research Support Grant, Yonsei University (2023)
- Rashtriya Uchchatar Shiksha Abhiyan Travel Fellowship, India (2019)
- National Eligibility Test (NET), AIR-96 (2014)
- National Eligibility Test (NET), AIR-75 (2013)
- National Scholarship (2010)

Teaching Experience

- Co-supervising M.Sc. and Ph.D. students, Yonsei University (2022–2025)
- Teaching Assistant: Engineering Geology, Jadavpur University (2018–2019)
- Co-supervising M.Sc. and Intern students, Jadavpur University (2016–2017)
- Teaching Assistant: Structural Geology Field Work, Jadavpur University (2015–2016)

Skills

Languages: English, Bengali, Hindi
Programming: MATLAB, FORTRAN90, Python, C++
Softwares/Codes: COMSOL Multiphysics, StagYY, GPlates, ParaView, ZMap, Global Mapper, GMT
Modelling: Finite Element Method (FEM), Finite Difference Method (FDM)

Synergistic Activities

- Reviewer for: Nature Communications, Communications Earth and Environment, Frontiers in Earth Science, GCubed, Geophysical Research Letters, Tectonophysics, Physics of Fluids, Geomorphology, Journal of Earth System Sciences, Geology, Minerals, Geosciences, Applied Sciences
- Outstanding Student Presentation Awards Judge: AGU (2023), JpGU (2024)
- Professional Affiliations: AGU, EGU, JpGU, SGTSGI

Referees

Prof. Nibir Mandal

Department of Geological Sciences, Jadavpur University, Kolkata, India

Ph.D. Supervisor, Email: nibir.mandal@jadavpuruniversity.in

Prof. Changyeol Lee

Department of Earth System Sciences, Yonsei University, Seoul, Republic of Korea

Former Postdoctoral Supervisor, Email: changyeol.lee@yonsei.ac.kr

Prof. Maëlis Arnould

Laboratoire de Géologie de Lyon Terre Planètes Environnement (LGL-TPE), Université Claude Bernard Lyon 1, France

MSCA Postdoctoral Host Supervisor, Email: maelis.arnould@univ-lyon1.fr

Prof. Maxim Ballmer

Department of Earth Sciences, Univ. College London (UCL), UK

MSCA Postdoctoral Supervisor, Email: m.ballmer@ucl.ac.uk